

Final Project: Audio Equalizer

Author: Soumil Iyer

Lab Partner: Student collaborator (name omitted for privacy)

Section: 012 (course staff name omitted for privacy)

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Abstract

This report explores the construction and performance of an audio equalizer circuit. The construction of the circuit will be explained by splitting it into four stages where each stage has a specific function. The main components used in the circuit construction were resistors, capacitors, inductors, potentiometers, and operational amplifiers. The purpose of this audio equalizer is to split, amplify, and then recombine an audio signal to achieve the desired sound from a speaker. In order to evaluate the performance of the circuit, each stage will be tested, and the performance will be compared to the lab requirements. The first stage of the audio equalizer circuit splits the audio signal. The second stage amplifies the split signals and allows the user to control their volume. The third stage recombines the initially split signal and also allows the user to control the volume of the recombined signal. The final stage sends the signal through an audio amplifier that amplifies the signal by a factor of 20 and outputs this signal to a speaker.

1. Objectives

The overall objective of the report is to explain and demonstrate a functioning audio equalizer. The audio equalizer circuit will be split into four stages and each stage will be explored and described in this report:

1.1 Signal input and filtering stage

This stage involves sending the input signal through three filters: a high pass filter, a band pass filter, and a low pass filter.

1.2 Signal adjustment stage

This stage will allow for the adjustment of amplitudes of the signals output by each of the three filters individually. This will be achieved using operational amplifiers and potentiometers.

1.3 Signal recombination and main volume control stage

This stage will recombine the split signals and allow for adjustment of the amplitude/volume of the recombined signal. This will be achieved using a summing amplifier and a potentiometer.

1.4 Signal power amplifying and output stage

This stage will amplify the signal output from the previous stage appropriately so that it can be safely outputted to the speaker. This will be done using an audio power amplifier.

2. Theory

2.1 Signal input and filtering stage

The circuit in this stage utilizes an RC high pass filter, an RLC band pass filter, and an RC low pass filter.

2.1.1 RC high pass filter

An RC high pass filter is essentially a frequency dependent voltage divider. It allows signals with frequencies higher than a certain cutoff frequency to pass

through while attenuating signals with frequencies lower than the cutoff frequency. The circuit consists of a resistor and a capacitor connected in series. At low frequencies, the reactance of the capacitor is very high, hence it behaves like an open circuit. This effectively blocks input signals until the cutoff frequency is reached. Above the cutoff frequency, the capacitor's reactance decreases which allows most of the input signals to pass through to the resistor. The V_{out} of an RC high pass filter is measured across the resistor. The cutoff frequency f_{c1} is related to the resistance R_1 and capacitance C_1 as follows:¹

$$f_{c1} = \frac{1}{2\pi R_1 C_1}$$

It is important to note that this equation only holds true for the -3dB point, which is what is what is desired in this report. Moreover, in an audio equalizer, this filter would allow only treble or high pitch signals to pass through.

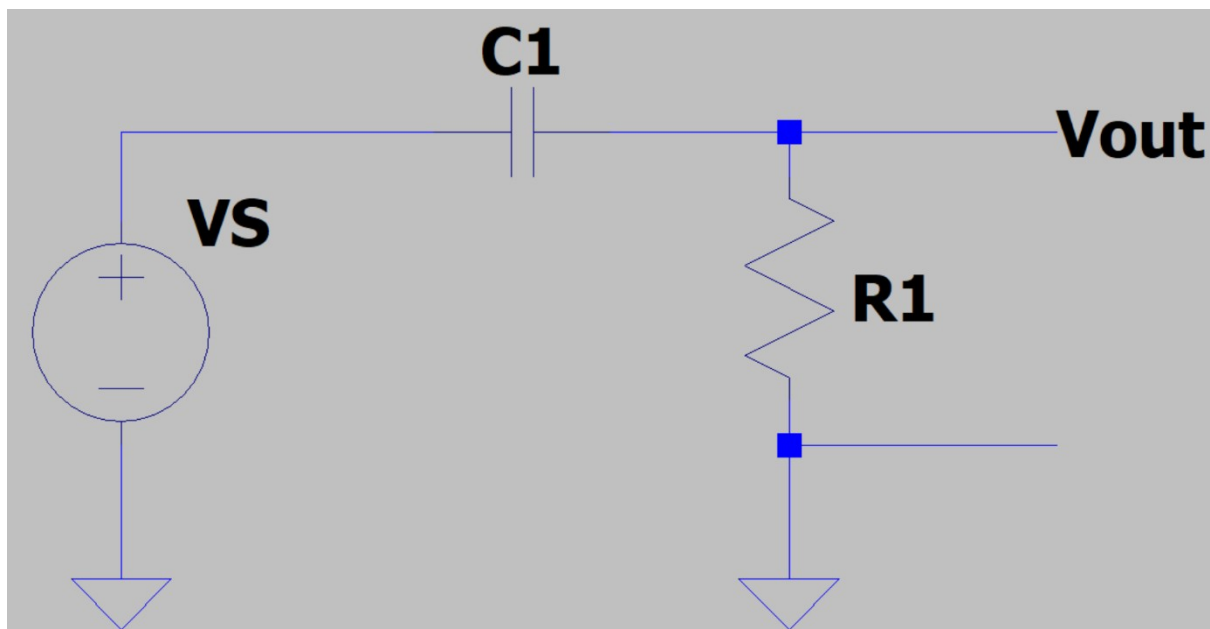


Figure 1: RC high pass filter circuit diagram

2.1.2 RLC band pass filter

An RLC band pass filter is a circuit that only allows signals with frequencies within a specific range to pass through while attenuating those outside of the range. This circuit consists of a resistor, an inductor, and a capacitor. The V_{out} of this circuit is measured across the resistor. The center frequency of an RLC band pass filter is at the center of the band (on a logarithmic scale). This center

¹ <https://www.electrical4u.com/high-pass-filter/>

frequency f_{c2} is the square root of the product of the starting frequency ω_a and ending frequency ω_b of the bandwidth. The center frequency f_{c2} can also be calculated using the impedance L_2 and capacitance C_2 as follows:²

$$f_{c2} = \sqrt{\omega_a \omega_b} = \frac{1}{2\pi \sqrt{L_2 C_2}}$$

Furthermore, the resistance R_2 is related to the starting and ending frequencies of the bandwidth along with impedance L_2 as follows:³

$$\omega_b - \omega_a = \frac{R_2}{L_2}$$

In an audio equalizer, this filter would allow only mid frequencies to pass through.

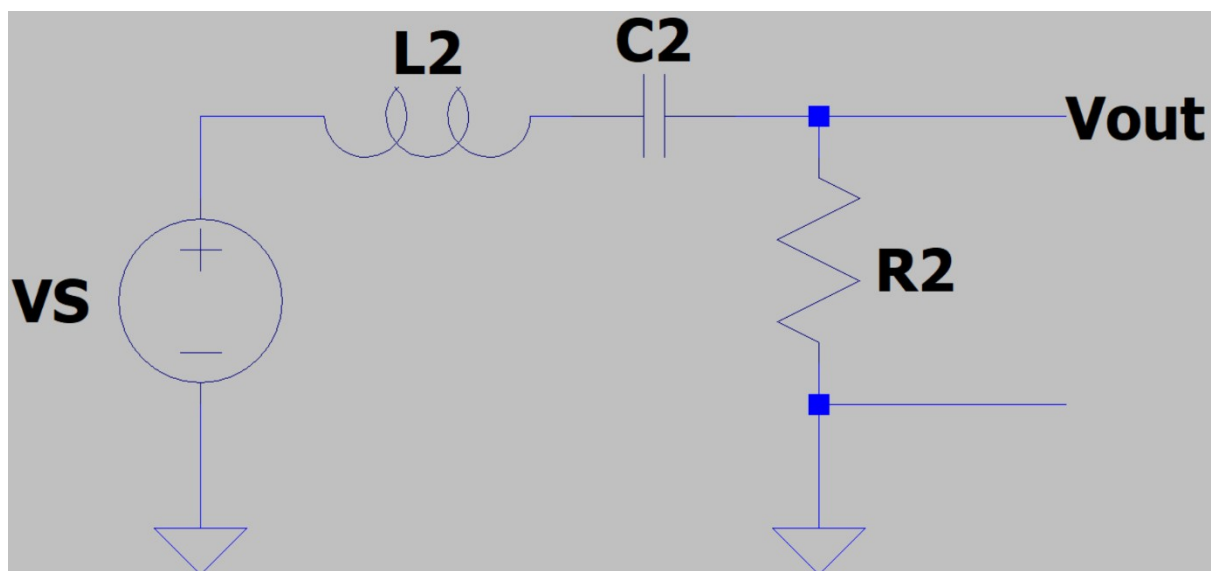


Figure 2: RLC band pass filter circuit diagram

2.1.3 RC low pass filter

An RC low pass filter is also a frequency dependent voltage divider. It only allows signals with frequencies lower than a certain cutoff frequency to pass through while attenuating signals with frequencies higher than the cutoff frequency. This circuit also consists of a resistor and a capacitor connected in

²<https://www.shahucollegelatur.org.in/Department/Studymaterial/sci/phy/Passive%20Band%20Pass%20RC%20Filter.pdf-converted.pdf>

³ <https://www.electrical4u.com/band-pass-filter/>

series. The V_{out} of an RC low pass filter is measured across the capacitor. The cutoff frequency f_{c3} is related to the resistance R_3 and capacitance C_3 as follows:⁴

$$f_{c3} = \frac{1}{2\pi R_3 C_3}$$

Please note, this formula is identical to that for the RC high pass filter.

Moreover, just like for the RC high pass filter, this equation only holds true for the -3dB point. Furthermore, in an audio equalizer, this filter would allow only bass or low range signals to pass through.

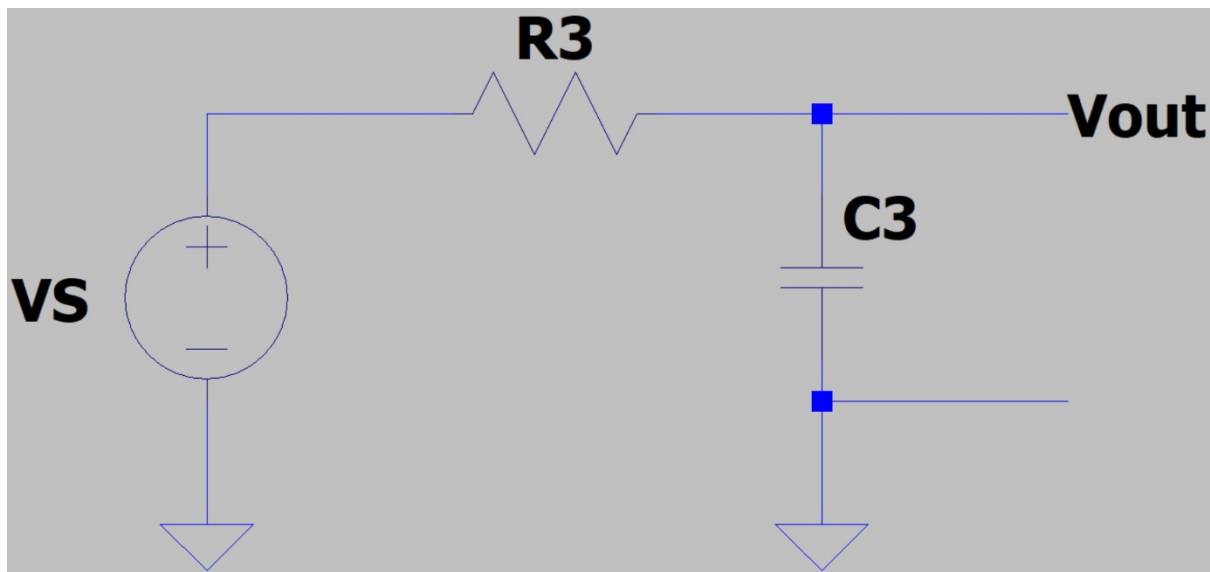


Figure 3: RC low pass filter circuit diagram

2.2 Signal adjustment stage

The circuit in this stage consists of three identical inverting amplifiers. The output from each of the three filters from the previous stage is individually input into three separate inverting amplifier circuits.

An inverting amplifier is a type of operational amplifier that results in an output that is 180° out of phase compared to the input. The voltage gain of an inverting amplifier is the negative ration between R_2 and R_1 ($\frac{v_{out}}{v_{in}} = \frac{-R_2}{R_1}$). Using a potentiometer as R_2 allows the gain to be varied.⁵

The signal from each of the filters from the previous stage will pass through one of the three inverting amplifiers and the potentiometer at R_2 will allow

⁴ <https://www.allaboutcircuits.com/technical-articles/low-pass-filter-tutorial-basics-passive-RC-filter/>

⁵ <https://www.electrical4u.com/inverting-amplifier/>

control of the “volume” of the output signal. When used in an audio equalizer, this would allow the user to control the volume of the treble, mid frequencies, and bass.

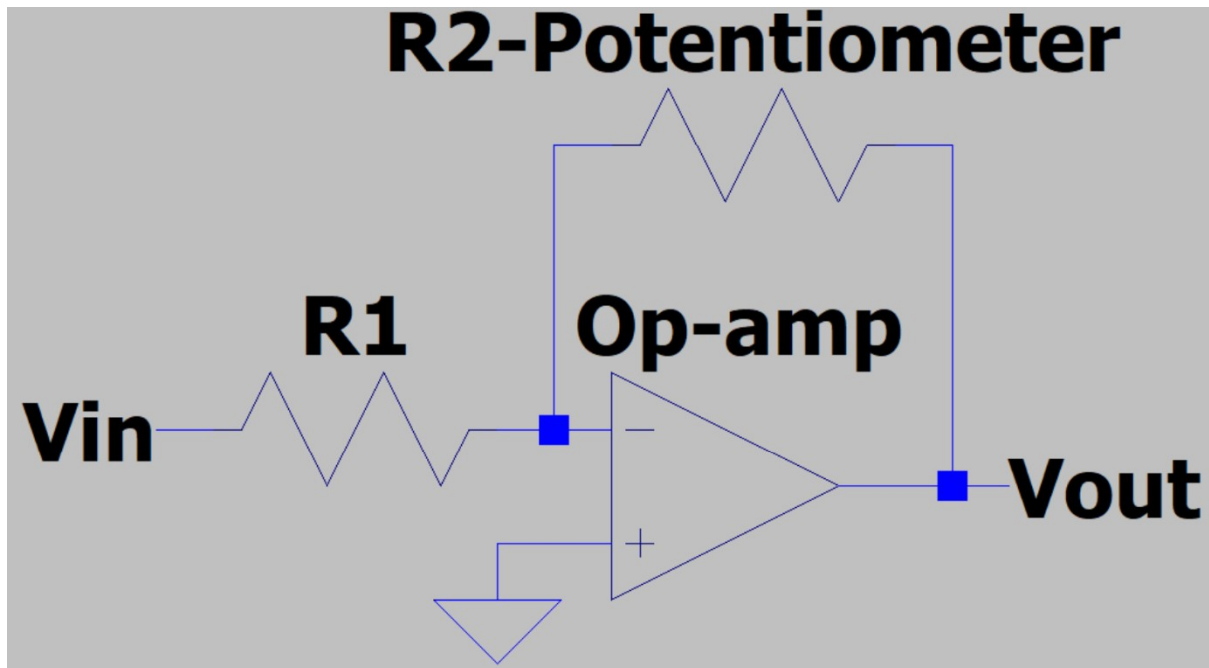


Figure 4: Inverting amplifier circuit diagram

2.3 Signal recombination and main volume control stage

This stage also uses an inverting amplifier circuit, the difference being that the three outputs from the previous stage will be input into a single amplifier. Hence, this inverting amplifier will have an input that is a recombination of the three signals that were initially split by the filters, hence the name “summing amplifier.” In an audio equalizer, this circuit would recombine the initially split treble, mid frequencies, and bass, and also allow the user to control the volume of the sound heard from the speakers connected later at the output stage.

The voltage gain of this amplifier is as follows:⁶

$$v_{out} = -R_F \cdot \left(\frac{v_{i1}}{R_1} + \frac{v_{i2}}{R_2} + \frac{v_{i3}}{R_3} \right)$$

⁶ <https://www.electrical4u.com/summing-amplifier/>

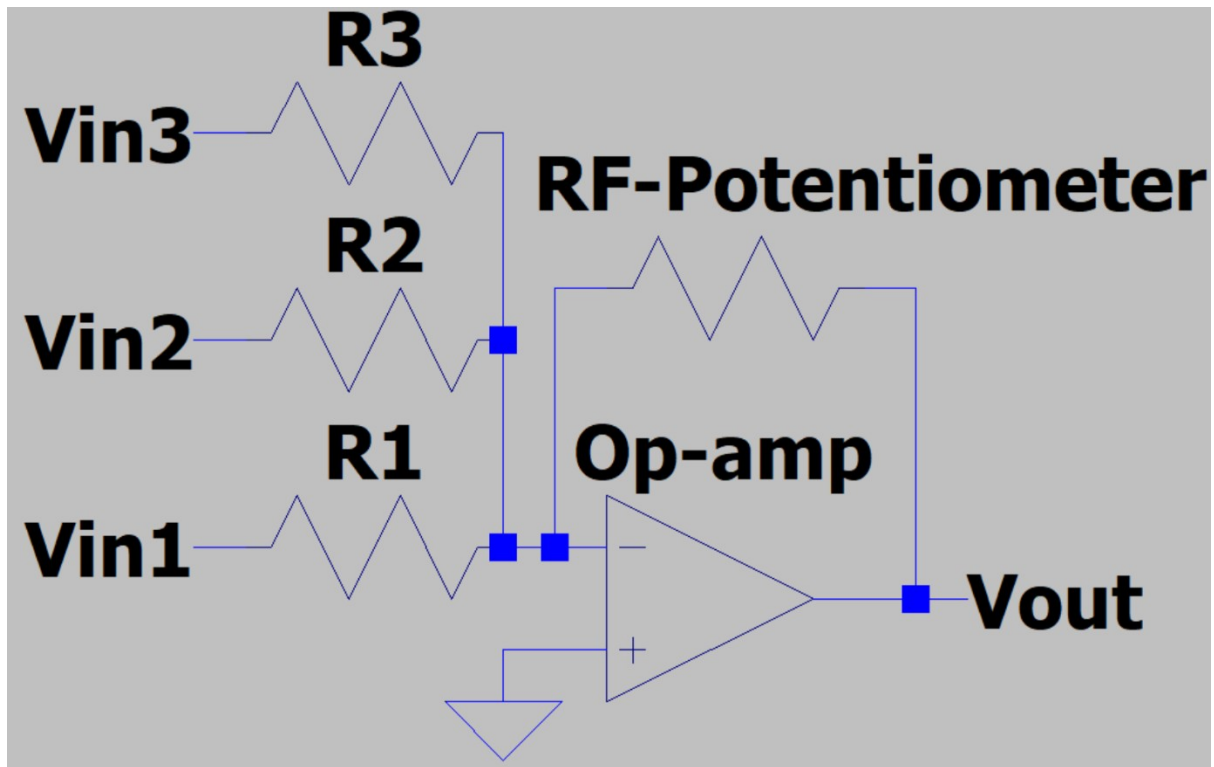


Figure 5: Summing amplifier circuit diagram

2.4 Signal power amplifying and output stage

The circuit in this stage also uses an amplifier, however the amplifier used here is specifically designed to be used as an audio power amplifier. It essentially amplifies low-power signals to a level that can drive a speaker. Its gain is internally set to 20 and that is the gain which would be used in the overall circuit.⁷ The output of this signal will go to the speaker and it will output sound that corresponds to the signal.

3. Procedure

3.1 Signal input and filtering stage

The circuit in Figure 6 was constructed which consists of an RC high pass filter (top), a band pass filter (middle), and an RC low pass filter (bottom). The formulas stated in 2.1.1, 2.1.2, and 2.1.3 respectively were used to calculate the component values.

⁷ <https://www.ti.com/lit/ds/symlink/lm386.pdf>

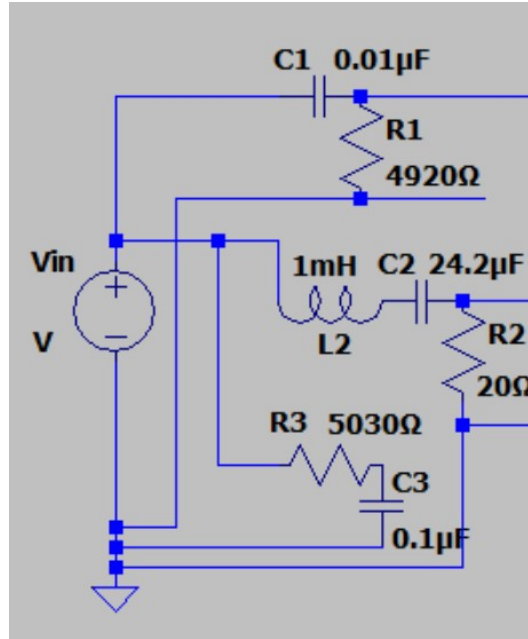


Figure 6: Signal input and filtering stage circuit diagram

3.1.1 RC high pass filter

In Figure 6, the RC high pass filter is at the top and the components in this filter are R_1 and C_1 . The requirements for this filter were having a -3dB cutoff of 3200Hz. The $0.01\mu\text{F}$ capacitor was chosen initially since there were a limited number to choose from. Based on this, the value for R_1 was calculated as follows:

$$f_{c1} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi R_1 \cdot 0.01 \times 10^{-6}} = 320 \text{ Hz}$$

$$R_1 = 4973.59 \Omega$$

The closest resistor combination possible was $4700\Omega + 220\Omega$. Hence, the resistance R_1 used in this circuit was 4920Ω .

3.1.2 RLC band pass filter

In Figure 6, the RLC band pass filter is at the middle and the components in this filter are R_2 , L_2 , and C_2 . The requirements for this filter were having a -3dB cutoff of 3200Hz. The 1mH inductor was chosen initially since that was the only inductor available. Based on this, the values for R_2 and C_2 were calculated as follows:

$$f_{c2} = \sqrt{\omega_a \omega_b} = \frac{1}{2\pi \sqrt{L_2 C_2}}$$

$$f_{c2} = \sqrt{320 \times 3200} = \frac{1}{2\pi \sqrt{0.001 \cdot C_2}}$$

$$C_2 = 24.7366 \mu F$$

The closes combination possible was $22\mu F + 2.2\mu F$, hence the capacitance C_2 used in this circuit was $24.2\mu F$.

$$\omega_b - \omega_a = \frac{R_2}{L_2}$$

$$2\pi(3200 - 320) = \frac{R_2}{0.001}$$

$$R_2 = 18.1 \Omega$$

The closest resistor combination possible was $10\Omega + 10\Omega$. Hence, the resistance R_2 used in this circuit was 20Ω .

3.1.3 RC low pass filter

In Figure 6, the RC low pass filter is at the bottom and the components in this filter are R_3 and C_3 . The requirements for this filter were having a -3dB cutoff of 320Hz. The $0.1\mu F$ capacitor was chosen initially since there were a limited number to choose from. Based on this, the value for R_3 was calculated as follows:

$$f_{c1} = \frac{1}{2\pi R_3 C_3} = \frac{1}{2\pi R_3 \cdot 0.1 \times 10^{-6}} = 3200 \text{ Hz}$$

$$R_1 = 4973.59 \Omega$$

The closest resistor combination possible was $4700\Omega + 330\Omega$. Hence, the resistance R_3 used in this circuit was 5030Ω .

3.2 Signal adjustment stage

The main component used in this stage of the circuit is the LM324⁸ operational amplifier. This component has four amplifiers within it. This stage of the circuit used three of those amplifiers. This circuit was constructed as shown in Figure 7. The inputs to each one of these inverting amplifiers is the output of each filter from the previous stage, i.e., the output of the RC high pass filter is the input for the first inverting amplifier and so on. The voltage gain needs to be controllable between 0 and -1. Hence, the components were chosen as follows:

⁸ <https://www.onsemi.com/pdf/datasheet/lm324-d.pdf>

R_{p1} , R_{p2} , R_{p3} are $10k\Omega$ potentiometers and R_4 , R_5 , R_6 are $10k\Omega$ resistors. Since the resistance of the potentiometer can be varied between $10k\Omega$ and 0Ω , this results in a variable gain between 0 and -1. The voltage supplied to the three amplifiers is +9V and -9V. The three inverting amplifiers are identical and the potentiometer dial allows the user to control the “volume” or treble, mid frequencies, and bass.

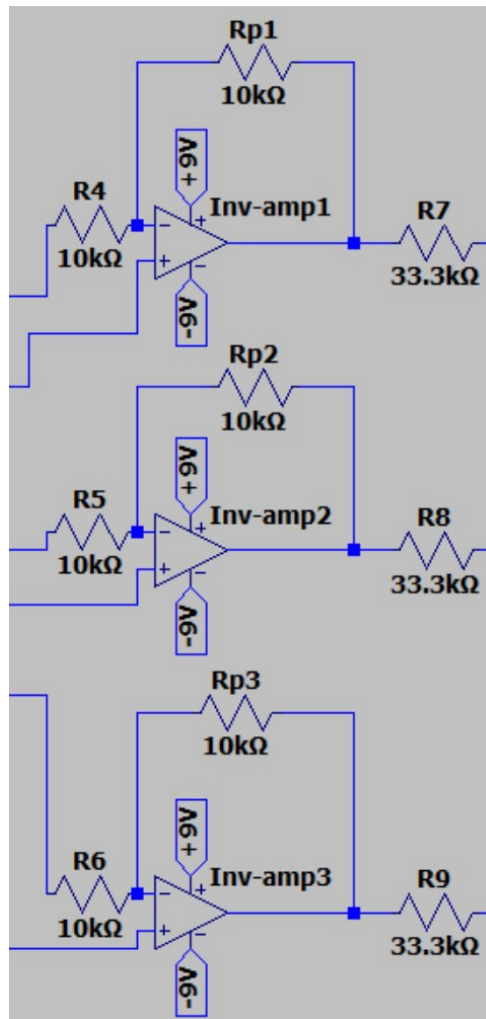


Figure 7: Signal adjustment stage circuit diagram

3.3 Signal recombination and main volume control stage

The circuit in this stage uses an LF356⁹ operational amplifier. This is a regular amplifier that was used as the summing amplifier in this circuit. This circuit was constructed as shown in Figure 8. In order to have a variable and controllable

⁹ <https://wp.optics.arizona.edu/mnofziger/wp-content/uploads/sites/31/2016/08/LF356.pdf>

gain between 0 and -1, the following components were chosen. R_7 , R_8 , and R_9 are the resistors between the output of the RC high pass filter, RLC band pass filter, and RC low pass filter, and the input of the summing amplifier respectively. R_{p4} is a 10k Ω potentiometer. In order to have a variable gain of 0 to -1, R_7 , R_8 , and R_9 were all chosen to be 33k Ω capacitors. With the combined resistance at the input of the summing amplifier being 11k Ω , the range of the gain is approximately between 0 and -1.1 which is close enough to the desired range of 0 to -1. This amplifier effectively recombines the initially split signals. The voltage supplied to the amplifier is +9V and -9V. The potentiometer dial allows the user to control the “volume” of the recombined signal.

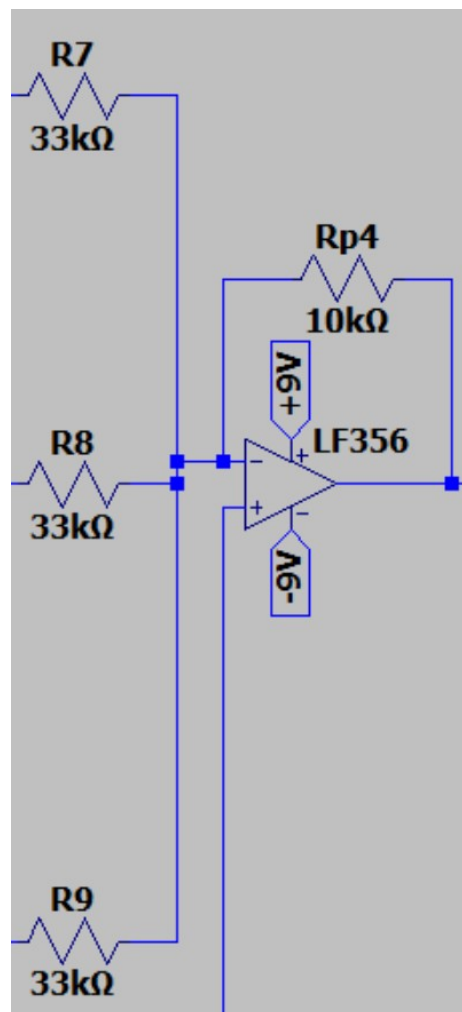


Figure 8: Signal recombination and main volume control stage

3.4 Signal power amplifying and output stage

The final stage of the audio equalizer uses the LM386 audio amplifier. This is an amplifier that is specifically designed to be used with low voltage audio signals.

The audio amplifier's input is the recombined output signal from the previous stage. The output terminal of the amplifier goes to the speaker and to a $0.05\mu\text{F}$ capacitor and 10Ω resistor connected in series which are grounded. This is the standard circuit setup for an LM386 audio amplifier in order to have a gain of 20.¹⁰ The voltage supplied to the three amplifiers is +9V.

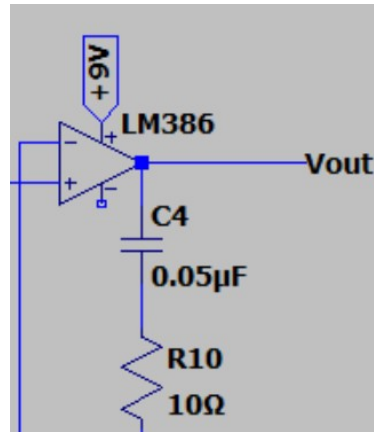


Figure 9: Signal power amplifying and output stage circuit diagram

3.5 Overall audio equalizer circuit

Figure 10 shows the combined audio equalizer circuit. This diagram puts together the circuits from all the four stages and results in a circuit that can output audio when the V_{out} terminal is connected to a speaker. The input V_{in} can either be a wave or an audio signal from a computer or any other device. Potentiometers R_{p1} , R_{p2} , R_{p3} can adjust the treble, mid frequencies, and bass respectively, and potentiometer R_{p4} can adjust the volume of sound output from the speaker.

¹⁰ <https://www.ti.com/lit/ds/symlink/lm386.pdf>

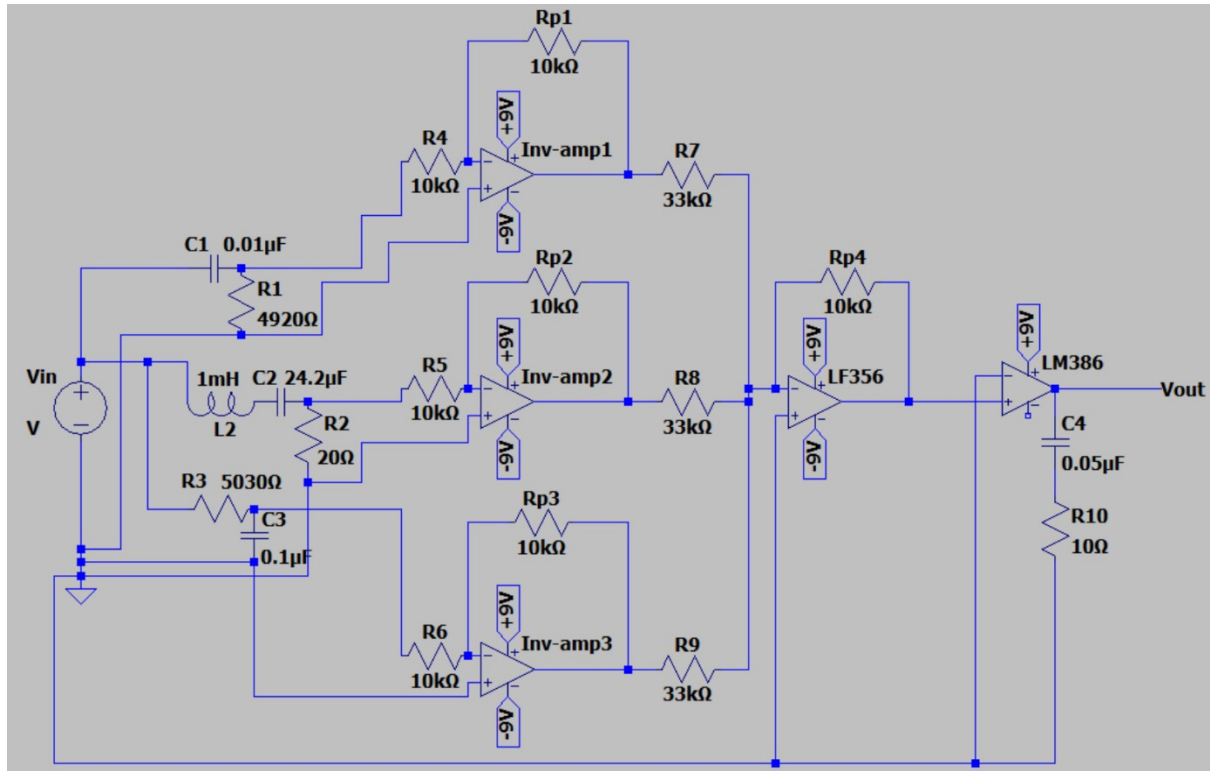


Figure 10: Overall audio equalizer circuit diagram

4. Results

The performance of the filters from the first stage in the actual circuit will be analyzed in below.

Figure 11 below shows the frequency response analysis (FRA) plot for the high pass filter. Since the cursors could not be exactly positioned at the -3dB point, two separate cursors were positioned before and after the -3dB point. Since the error tolerance is $\pm 10\%$ with respect to 3.2kHz, the upper limit of the acceptable -3dB cutoff frequency is 3.52kHz. The values of cutoff frequencies at -3.20dB and -2.89dB are 3.311kHz and 3.548kHz respectively. Since the value of the -3dB cutoff frequency is going to be closer to that at -2.89dB, the cutoff frequency at -3dB can be safely considered to be within the tolerance.

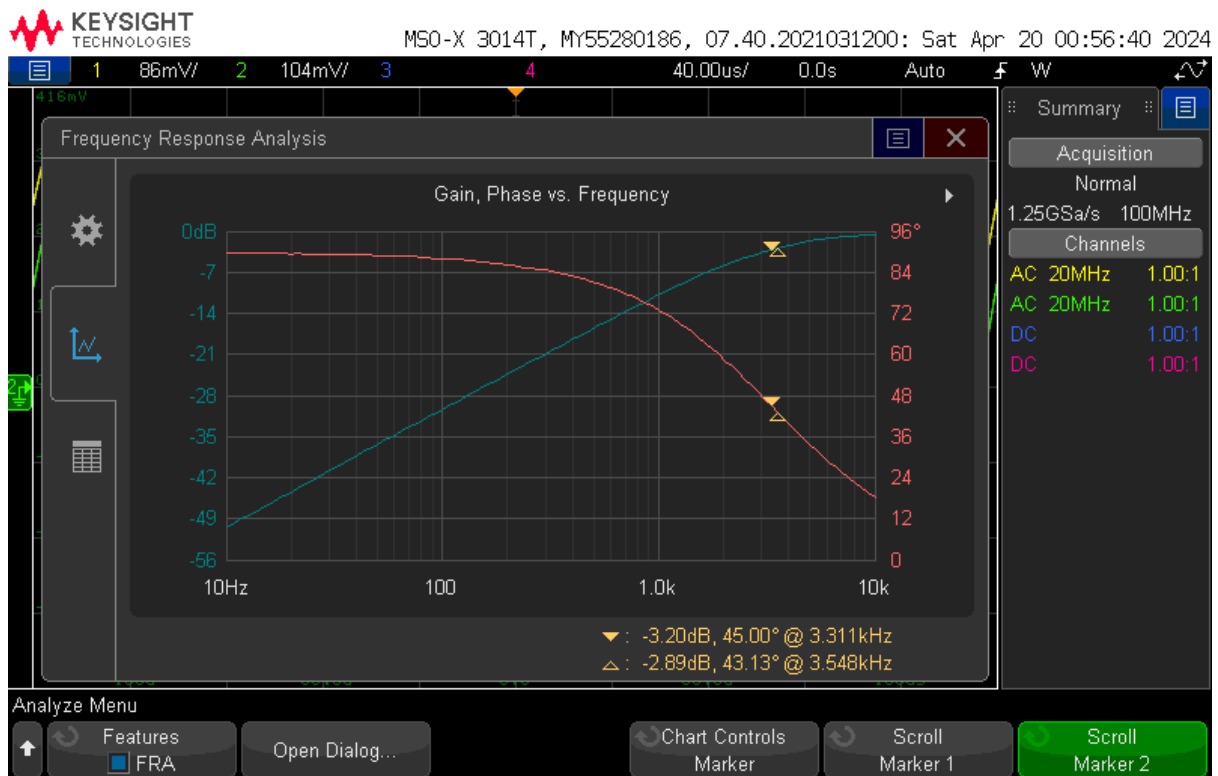


Figure 11: RC high pass filter frequency response analysis plot

Figure 12 below shows the frequency response analysis (FRA) plot for the band pass filter. Again, the cursors could not be exactly positioned at the -3dB points, however they are closer in this plot. According to the FRA plot, the bandwidth of the RLC band pass filter is 446.7Hz – 3.311kHz. While the latter is within the upper limit of 3.52kHz, the former is not within the upper limit of 352Hz. However, since the filters do not entirely block out frequencies that are not in the bandwidth but rather attenuate them, for the purpose of the audio equalizer, this should not create an issue.

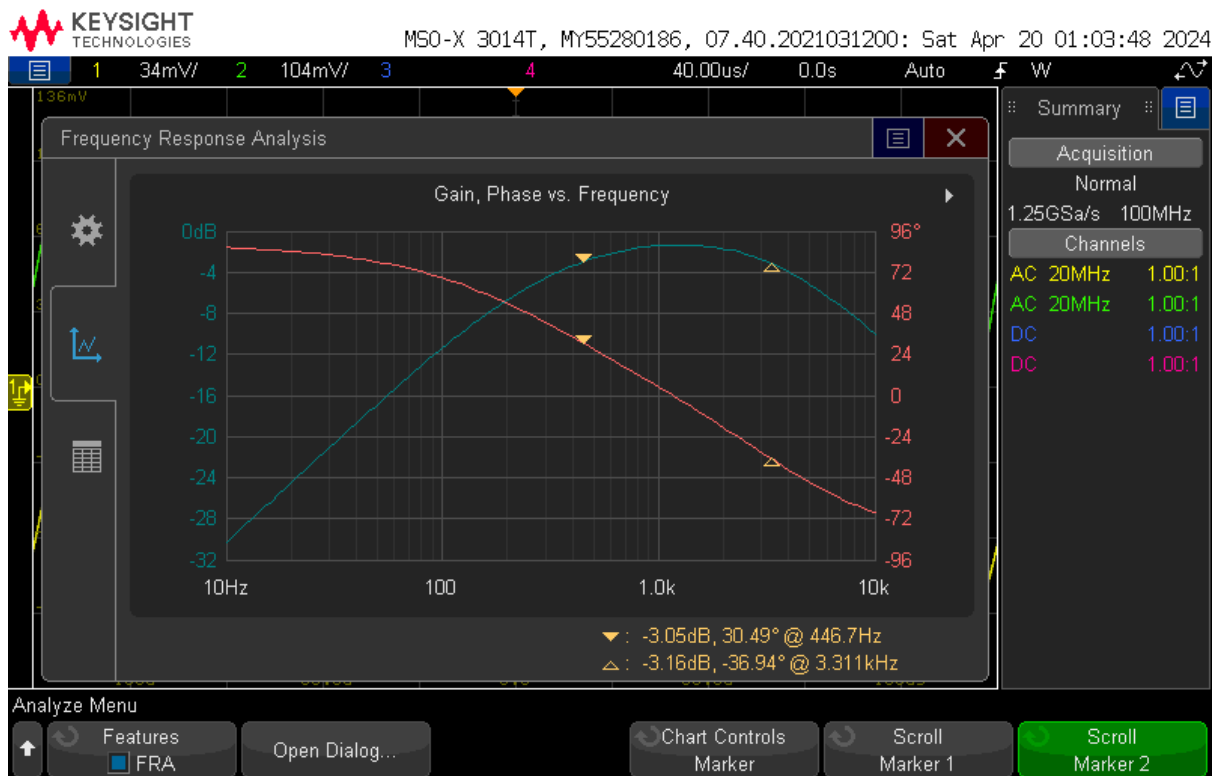


Figure 12: RLC band pass filter frequency response analysis plot

Figure 13 below shows the frequency response analysis (FRA) plot for the low pass filter. Again, the cursors could not be positioned exactly at the -3dB point. Since the range of acceptable cutoff frequencies at -3dB is 288Hz – 352Hz, and the cutoff frequencies at -2.91dB and -3.19dB are 316.2Hz and 338.8Hz, then the cutoff frequency at -3dB is certainly within the acceptable range.

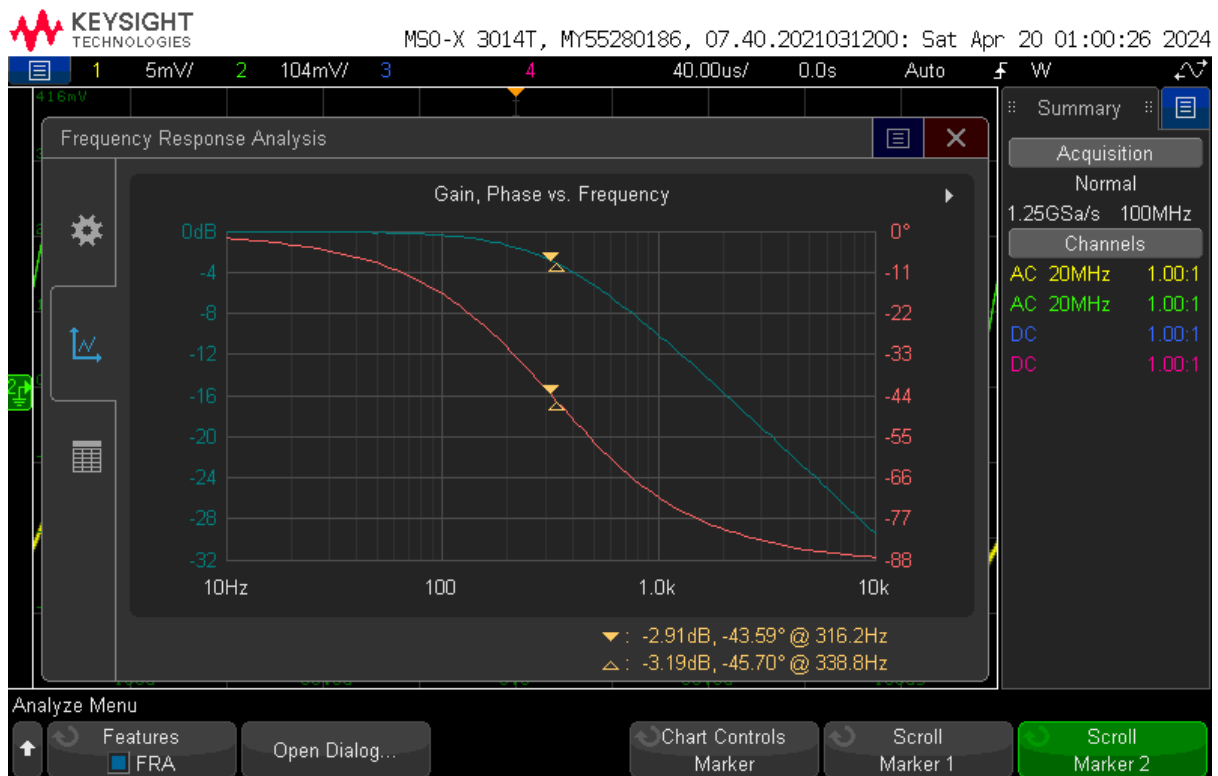


Figure 13: RC low pass filter frequency response analysis plot

Next, the maximum output of the entire circuit right before the audio amplifier (fourth stage) was measured. The output was maximized by turning all the four potentiometer dials to their maximum resistance: 10k Ω . This means that the gain was maximum for the four amplifiers that used potentiometers. With these settings, the plot in Figure 14 below was derived.

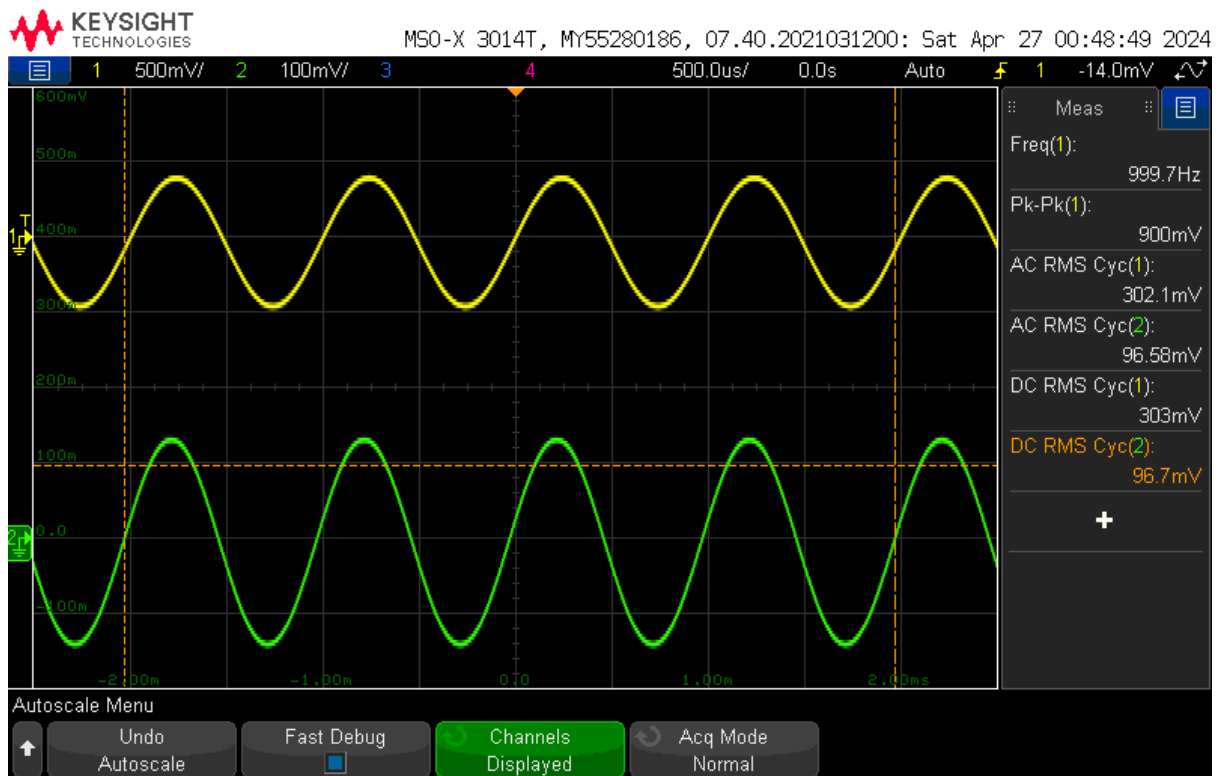


Figure 14: Maximum V_{amp} (output of third stage – summing amplifier)

The output before the audio amplifier was also measured with all the potentiometer dials turned down to their lowest setting, i.e., 0Ω . With these settings, not signal or a very small signal should be expected. This is shown in the plot in Figure 15.

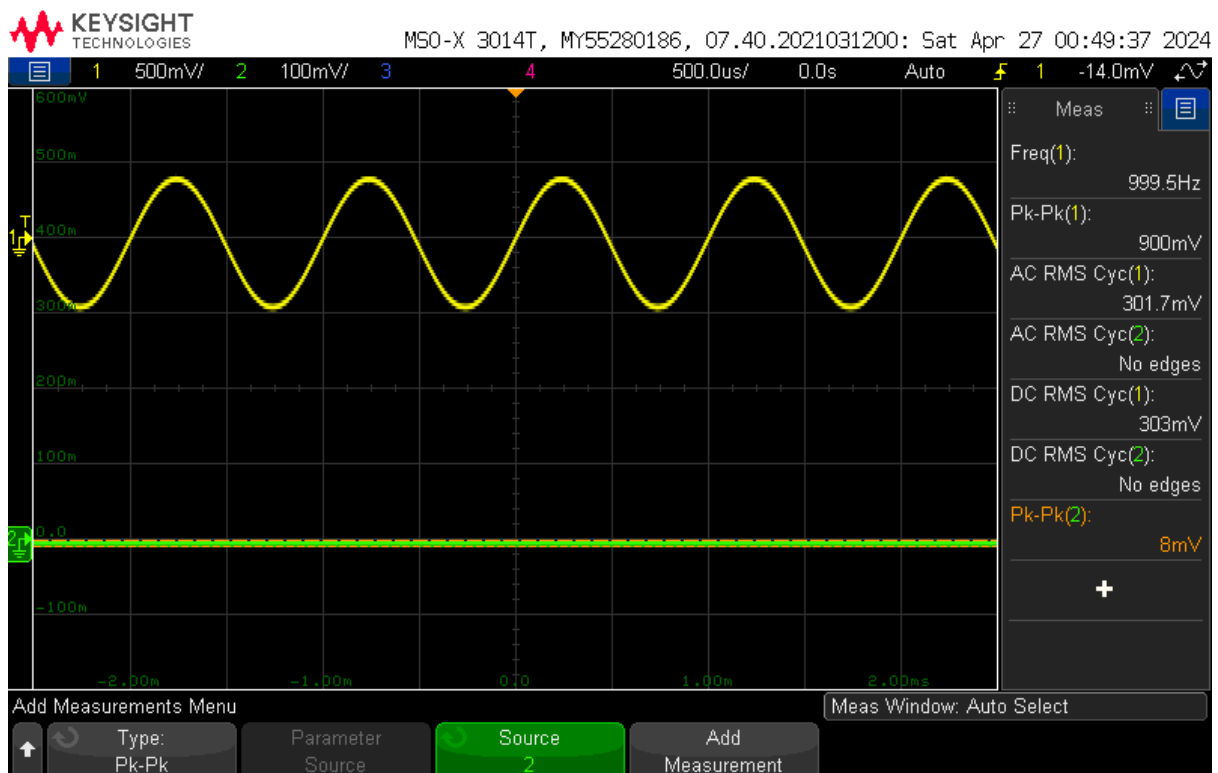


Figure 15: Minimum V_{amp} (output of third stage – summing amplifier)

Table 1: Maximum and minimum V_{amp} measurements

	Measured (mV)	Expected (mV)	% error
Maximum V_{amp} RMS	96.7	100	3.30%
Minimum V_{amp} RMS	0	<15	-

Figure 16 shows the overall frequency response analysis of the output of the entire audio equalizer circuit. The blue curve needs to be mostly smooth and flat for clear and uninterrupted audio, which it is. This is also supported by the fact that the audio output from the speaker that is connected to the actual circuit was very clear and uninterrupted.

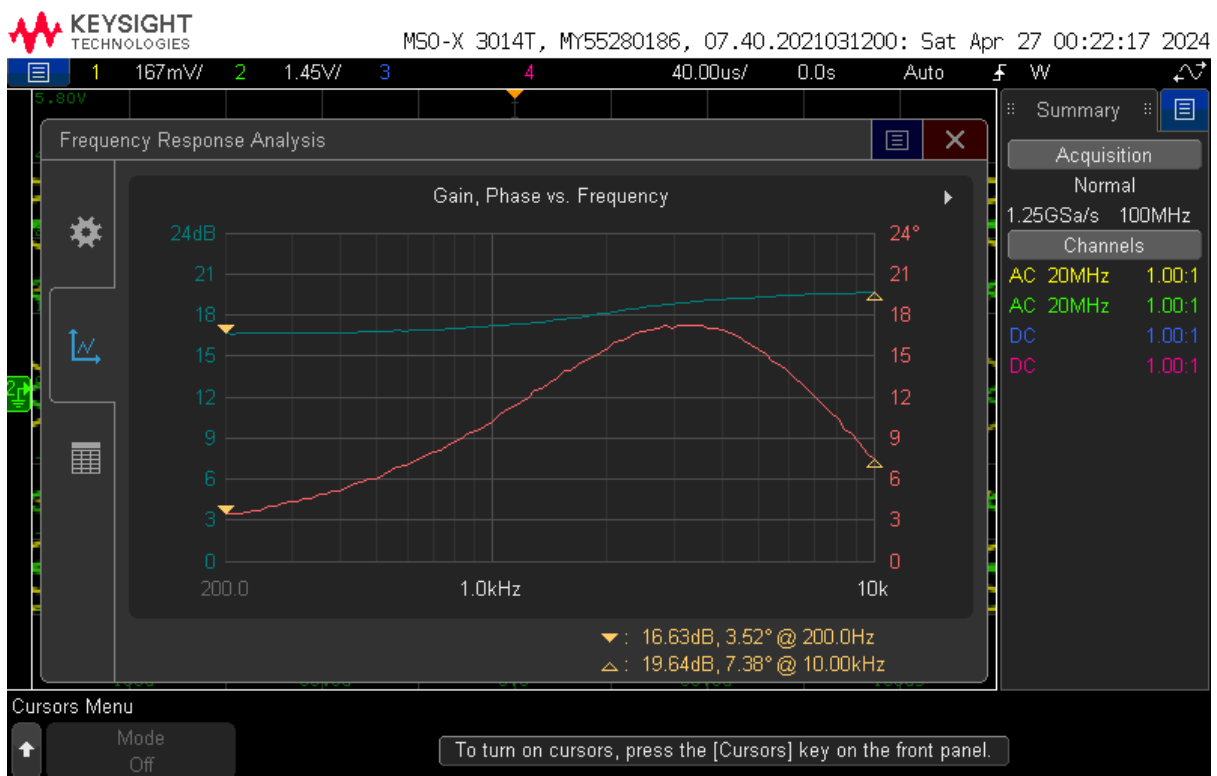


Figure 16: Frequency response analysis of overall circuit output

Figure 17 below shows the RMS voltage output of the audio equalizer. Based on this value, and the fact that the speakers resistance is 8Ω , the power output can be calculated as follows:

$$P_{out(RMS)} = \frac{2.2^2}{8} = 605 \text{ mW}$$

The expected power output should be a value greater than 400mW. Since the measured power output of 605mW is clearly greater, the audio equalizer circuit functions optimally as intended.

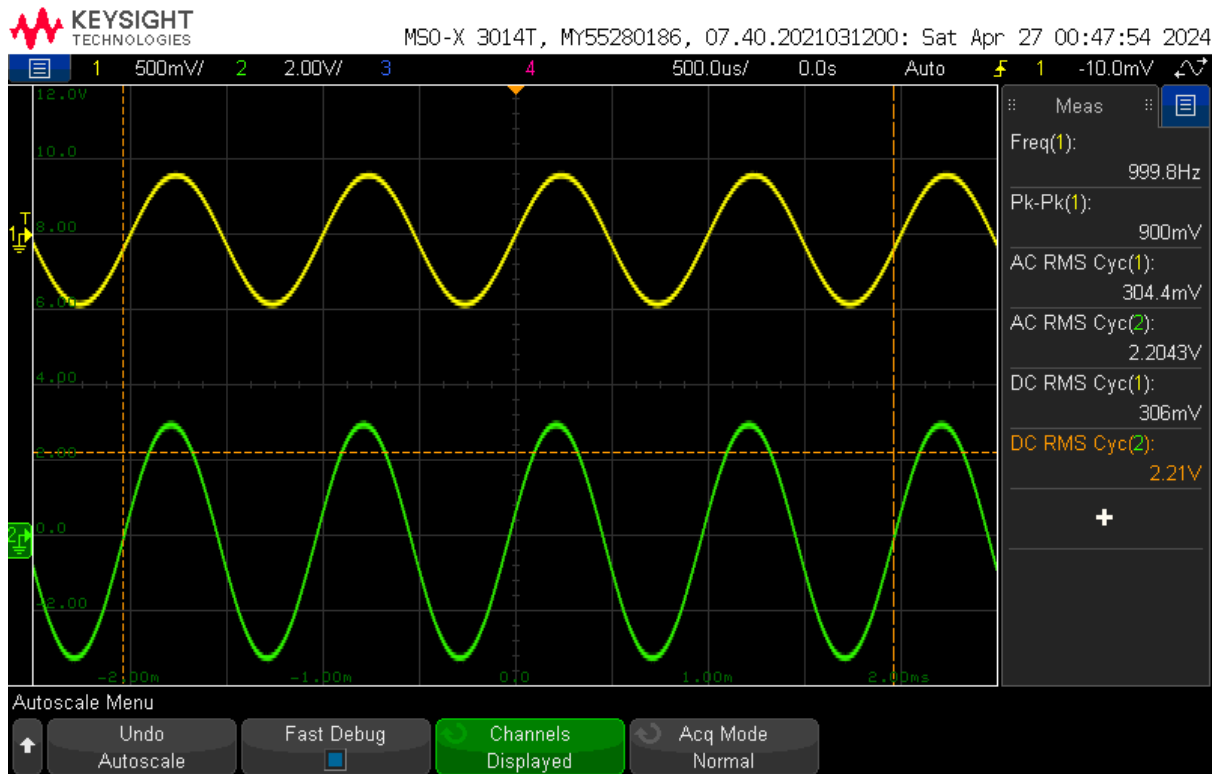


Figure 16: $V_{out(RMS)}$ of overall circuit

Improvements to the overall audio equalizer circuit design

In order to improve the performance of the audio equalizer, the following changes can be made:

- Using the exact resistances and capacitances calculated at each stage rather than using components that have a close enough resistance and capacitance.
- Using minimal and shorter wiring where possible to reduce the overall additional resistance in the circuit due to wires.

5. Conclusion

The audio equalizer circuit can be split into four stages where each stage serves a specific purpose. The first stage is the signal input and filtering stage where the input signal is sent through three filters: RC high pass, RLC band pass, and RC low pass filters. This splits the signal into trebles, mid frequencies and bass signals. Each of the three signals is then sent through an inverting amplifier which allows the “volume” of each signal to be varied using a potentiometer. These signals are then recombined by the summing amplifier and this inverting amplifier allows for volume control of the recombined audio signal. This recombined signal then goes through an audio amplifier which then outputs the signal to a speaker.

The total power output of the audio equalizer was measured to be 605mW, which is appropriate for the kind of speakers this audio equalizer is designed to be used with.

Overall, this is an extremely useful circuit that makes use of operational amplifiers, resistors, capacitors, and inductors in order to split, amplify, and recombine an audio signal to achieve the desired sound from a speaker.